



The Queue as Leading Indicator: Coffee-Cart Waiting Times and Global-Ranking Trajectory

Concourse queue telemetry as an anticipatory signal for international ranking position.

Casimir Beng · Osei Vandermeer · Anneke Tolan — School of Continuous Improvement

Motivation

Ranking position moves slowly and is read late — often a year after the activity that drove it. Little is known about whether everyday campus telemetry anticipates these moves. Does the coffee queue know first?

Research questions

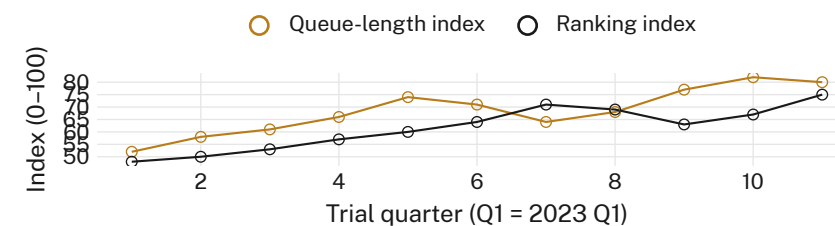
- Do coffee-cart queue lengths lead global ranking position?
- Is the lead stable enough to steer anticipatory strategy?
- Morning or afternoon queue — which carries the signal?

Study design

- 9 carts · overhead depth sensors · 5-min sampling · 24 months (2023–2025)
- De-identified queue index cross-correlated against monthly ranking; lag at peak cross-correlation
- Lag window and thresholds pre-registered; none adjusted post hoc

Findings

The queue-length index led ranking position by two quarters (peak cross-correlation $r = 0.78$). The lead held for both morning and afternoon queues, strongest at the carts nearest the senior common room.



Queue-length index (gold) led the ranking index (ink) by two quarters, 2023–2025: the March 2024 queue peak preceded the September 2024 ranking rise ($n = 9$ carts; peak cross-correlation $r = 0.78$).

Implications

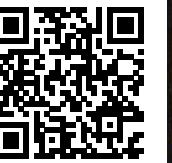
Results are consistent with queue telemetry acting as an early, low-cost proxy for the activity ranking bodies later reward — correlational, not causal, but stable across the trial. Next: a closed-loop pilot steering a fixed share of discretionary grant strategy from the live queue index, with no manual override.

Selected references

1. Kendall, D. G. (1953). Stochastic Processes Occurring in the Theory of Queues and their Analysis by the Method of the Imbedded Markov Chain, *The Annals of Mathematical Statistics*, 24(3), 338–354. doi:10.1214/aoms/1177728975
2. Espeland, W. N. & Sauder, M. (2007). Rankings and Reactivity: How Public Measures Recreate Social Worlds, *American Journal of Sociology*, 113(1), 1–40. doi:10.1086/517897
3. Billaut, J., Bouyssou, D., & Vincke, P. (2010). Should you believe in the Shanghai ranking?, *Scientometrics*, 84(1), 237–263. doi:10.1007/s11192-009-0115-x
4. Fleming, T. R. & DeMets, D. L. (1996). Surrogate End Points in Clinical Trials: Are We Being Misled?, *Annals of Internal Medicine*, 125(7), 605–613. doi:10.7326/0003-4819-125-7-199610010-00011
5. Eagle, N. & Pentland, A. (2006). Reality mining: sensing complex social systems, *Personal and Ubiquitous Computing*, 10(4), 255–268. doi:10.1007/s00779-005-0046-3

De-identified queue headcounts only; no faces or transactions retained. Supported by a School of Continuous Improvement seed grant.

Office of Research Outputs · slop.university · doi:10.5555/slop.sn9kzr



“Queue length led every ranking movement we observed, by a stable two quarters.”

Mean concourse queue-length index vs international ranking, 2023–2025 ($n = 9$ carts).